

Deep Learning the Anatomy of Growth: A Comparative CNN Study for Bone Age Prediction

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Abstract—Bone age estimation from pediatric hand radiographs is a clinically important task that benefits from automated, reproducible analysis enabled by modern deep learning. In this report, convolutional neural networks (CNNs) are investigated for bone age regression on the RSNA Pediatric Bone Age dataset, with an emphasis on how architectural design and training choices affect both accuracy and computational cost. Four lightweight, modified CNN families are implemented and compared: a VGG-inspired baseline CNN, a ResNet-inspired CNN with skip connections (SkipCon), an Inception-V4-inspired CNN (Inception), and an Inception-ResNet-V2-inspired CNN (InSkipCon). In addition, an ablation study evaluates the impact of CLAHE contrast enhancement, data augmentation, and the inclusion of gender as auxiliary metadata.

Results show that the best configuration is obtained with `Augment=False`, `CLAHE=True`, and `Gender=True`, where the InSkipCon model achieves a test MAE of 6.74 months (test RMSE 8.94 months). Overall, gender input consistently improves performance, CLAHE is generally beneficial, and augmentation does not provide reliable gains in this setting. The provided comparisons and implementation choices form a reproducible baseline for future work on accuracy-efficiency trade-offs in bone age prediction.

Index Terms—Bone Age Prediction, RSNA, Convolutional Neural Networks, Residual Connections, Inception Networks, CLAHE Enhancement

I. INTRODUCTION

The growing availability of medical imaging data and advances in deep learning have significantly accelerated the development of computer vision systems for healthcare. These models can now perform complex recognition tasks directly from images, reducing human workload and improving consistency in diagnostic processes. One of the clinically relevant applications of this technology is *pediatric bone age prediction*, where convolutional neural networks (CNNs) are used to estimate skeletal maturity from hand radiographs.

In pediatric endocrinology, bone age is one of the most reliable biological indicators of growth and maturation, providing an objective measure of skeletal development from birth through adolescence. The most established approaches, the *Greulich-Pyle (GP)* and *Tanner-Whitehouse (TW)* methods, rely on manual comparison of a child's hand and wrist radiograph to reference atlases [1]. The GP method remains the most commonly used due to its simplicity and speed, but it suffers from observer variability and limited generalizability beyond the original reference population. In contrast, the TW method evaluates 20 predefined regions of interest in

the hand and assigns maturity scores, offering higher reproducibility but at the cost of longer evaluation time and greater complexity [2]. Other traditional systems, such as the *Gilsanz-Ratib* atlases, provide refined or digital alternatives but remain dependent on manual interpretation [3]. To overcome these limitations, computer-aided systems like *CASAS* and *BoneXpert* have been introduced, automating feature extraction and reducing human error. However, these tools can be sensitive to image quality and still rely on handcrafted rules [4]. The recent introduction of deep learning-based models, particularly convolutional neural networks (CNNs), has advanced bone age assessment further by learning features directly from raw images. Such methods, including those evaluated in the *RSNA Pediatric Bone Age Machine Learning Challenge (2017)*, demonstrated that deep models can achieve expert-level precision while providing faster, more reproducible, and scalable assessment compared to traditional atlas-based techniques [4] [5].

The RSNA challenge represents a milestone in the application of deep learning to medical imaging, providing a large, publicly available dataset of pediatric hand radiographs and a standardized benchmark for model development. The competition encouraged participants to design deep learning models capable of accurately predicting skeletal maturity from images acquired at two different hospitals [6], effectively establishing the dataset as a reference point for subsequent research in this field and serves as the basis for this project.

The motivation behind this work is to explore and understand how different convolutional neural network (CNN) designs perform on the bone age prediction problem. The goal is to analyze how variations in architectural complexity and the integration of additional information influence regression accuracy and computational efficiency. To this end, four CNN-based models are implemented and evaluated under a common training and preprocessing framework: a baseline convolutional neural network, a skip-connection-enhanced CNN, an Inception-based CNN, and a hybrid Inception-Skip Connection architecture. All models operate directly on hand radiographs and optionally incorporate patient gender as additional information through a learned embedding, enabling the assessment of its contribution to skeletal age estimation. Through their systematic comparison, the study aims to provide practical insight into CNN behavior for bone age regression and to establish a clear, reproducible pipeline for medical image based age prediction.

This report is structured as follows. In Section II, the related work on bone age prediction and existing CNN-based

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approaches are reviewed. In Section III, the data processing pipeline and preprocessing steps are described. Section IV presents the dataset characteristics. In Section V, the learning framework, including the four proposed CNN architectures, is detailed. Section VI summarizes the results and discussions, and concluding remarks are provided in Section VII.

II. RELATED WORK

The development of automated bone age assessment methods has evolved from manual atlas-based techniques toward data-driven deep learning frameworks. Early approaches aimed to replace the Greulich-Pyle (GP) and Tanner-Whitehouse (TW) standards with computer-aided systems in order to reduce observer variability and improve reproducibility [4]. With the increasing availability of large annotated datasets, convolutional neural networks (CNNs) have become the dominant paradigm for learning discriminative representations directly from hand radiographs and performing skeletal age assessment.

A major milestone in this transition was the introduction of the RSNA Pediatric Bone Age Machine Learning Challenge, which provided a large-scale, multi-institutional dataset and established a standardized benchmark for performance comparison [5], [6]. Several early deep learning baselines demonstrated that CNN-based models could achieve expert-level accuracy without relying on handcrafted features. In particular, Spampinato et al. proposed *BoNet*, one of the first CNN architectures specifically designed for bone age estimation from hand X-rays [7]. Subsequently, Iglovikov et al. employed a *VGGNet*-based model combined with preprocessing steps such as hand segmentation and contrast normalization, providing an influential open-source baseline on the RSNA dataset [8]. These studies demonstrated the effectiveness of convolutional feature extraction for skeletal maturity assessment. Building on these foundations, later works explored more advanced CNN architectures. Several studies incorporated patient gender as additional input, showing that demographic information can improve regression accuracy [9], [10].

Residual connections, originally introduced in the ResNet architecture, were shown to improve training stability and generalization by facilitating gradient propagation in deep networks [11]. In parallel, Inception-style architectures enabled multi-scale feature extraction through parallel convolutional branches, allowing both local and global patterns to be captured efficiently [12]. These architectural design principles have since been widely adopted in bone age prediction models and medical image regression tasks, demonstrating favorable trade-offs between prediction accuracy and computational complexity.

More recent research has focused on region-aware and explainable frameworks. For example, Li et al. proposed a cascaded model that performs annotation-free region localization and gender-assisted bone age regression using CNN backbones such as Inception and ResNet [13]. Similarly, Andleeb et al. employed a DenseNet-based architecture with

activation-based explainability techniques to visualize anatomically relevant regions contributing to bone age estimation [14]. While these methods highlight the benefits of region-focused processing and interpretability, they often rely on complex pipelines or pretrained networks.

In contrast, the present work focuses on a systematic comparison of lightweight CNN architectures trained in an end-to-end fashion. Inspired by the design principles of VGG [15], ResNet [11], Inception-V4, and Inception-ResNet-V2 [12], simplified and lightweight variants of these architectures are evaluated under a unified training and preprocessing framework. By emphasizing architectural design choices and additional gender information integration, this study aims to assess how different CNN paradigms influence accuracy and efficiency in bone age regression.

III. PROCESSING PIPELINE

The proposed system follows a unified and modular processing pipeline designed to train and evaluate multiple convolutional neural network (CNN) architectures for pediatric bone age prediction from hand radiographs. A high-level overview of the pipeline is illustrated in Fig. 1. The pipeline is structured to ensure consistency across models and to isolate architectural design choices as the primary factor influencing performance.

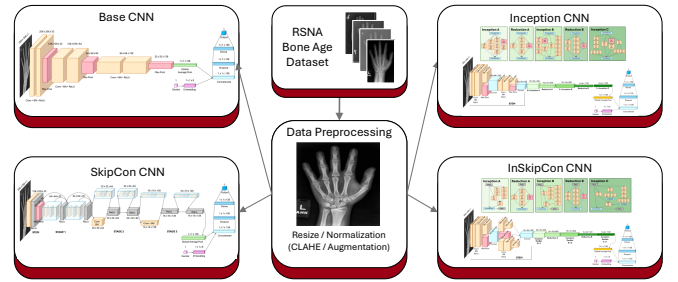


Fig. 1: Processing Pipeline

The input to the system consists of pediatric hand X-ray images from the RSNA Pediatric Bone Age dataset, accompanied by patient gender information. As an initial preprocessing step, all radiographs are resized to a fixed spatial resolution to ensure uniformity across the dataset. Pixel intensities are subsequently normalized using z-score normalization to stabilize training and reduce sensitivity to global illumination differences. In addition to this base preprocessing, optional image enhancement and data augmentation techniques are evaluated as part of an ablation study. Specifically, contrast-limited adaptive histogram equalization (CLAHE) [16] is optionally applied to enhance local contrast and improve the visibility of bone structures, while geometric data augmentation is selectively enabled during training to assess its impact on generalization performance. These preprocessing options are controlled via configuration and do not alter the core structure of the processing pipeline.

After preprocessing, the images are passed to a set of CNN-based regression models, each trained independently under

the same data splits, optimization strategy, and evaluation metrics. The considered architectures include a baseline convolutional network and three advanced variants incorporating skip connections and multiscale processing. All models operate directly on full hand radiographs and are trained from scratch without the use of pretrained weights. Patient gender is incorporated as auxiliary information through a learned embedding, which is concatenated with the image derived feature representation prior to the regression layers.

Model outputs correspond to a continuous prediction of skeletal age expressed in months, and the learning task is formulated as a regression problem. Model performance is evaluated using the mean absolute error (MAE), which provides a clinically interpretable measure of prediction accuracy. To improve training stability and prevent overfitting, early stopping and an adaptive learning-rate scheduler are employed across all experiments.

IV. SIGNALS AND FEATURES

This section describes the input signals used in the study, their formatting, and the preprocessing strategies applied prior to model training.

Dataset and Input Formatting. The experiments are conducted using the *RSNA Pediatric Bone Age* dataset, which consists of left-hand radiographs acquired from pediatric patients at two clinical institutions. Left-hand images are used in accordance with standard clinical practice, as bone age assessment traditionally relies on the non-dominant hand to reduce confounding effects related to dominant-hand overuse [13]. This practice aligns with the Greulich-Pyle (GP) and Tanner-Whitehouse (TW) standards, where left-hand radiographs are the clinical norm.

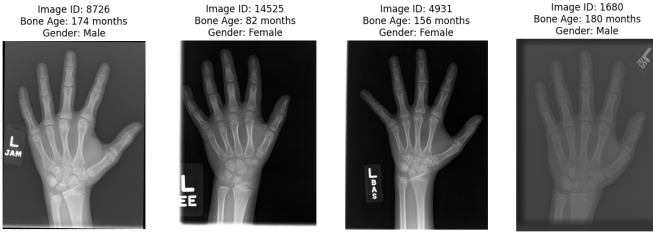


Fig. 2: Sample images from the dataset

Each radiograph is provided as a PNG image and is associated with a corresponding entry in a CSV file containing:

- the *bone age in months*, used as the regression target,
- the patient's *gender* (male/female).

Radiographs are represented as single-channel grayscale images, reflecting their monochromatic nature and reducing computational complexity by limiting the number of learnable parameters in early convolutional layers. Representative examples from the dataset are shown in Fig. 2.

Preprocessing of Radiographic Signals. Prior to model training, all radiographs undergo a base preprocessing procedure designed to standardize the input format and stabilize learning. First, images are resized to a fixed spatial resolution

of $(256 \times 256 \times 1)$, producing uniform input tensors across all experiments. Pixel intensities are then standardized using z-score normalization,

$$I_{\text{norm}} = \frac{I - \mu}{\sigma},$$

where μ and σ denote the mean and standard deviation of pixel intensities. This normalization is widely adopted in medical imaging, as radiological modalities (e.g., X-ray, MRI, CT) often exhibit large variations in brightness, exposure, and acquisition conditions across machines and hospitals [17]. By standardizing intensity distributions, z-score normalization helps mitigate these global illumination differences, leading to more consistent feature extraction across samples.

In addition to this base preprocessing, optional image enhancement and data augmentation techniques are evaluated as part of an ablation study. Specifically, contrast-limited adaptive histogram equalization (CLAHE) [16] may be applied to enhance local contrast and improve the visibility of bone boundaries and growth plates. The impact of CLAHE on bone age prediction has been investigated in prior studies [5], [10], [18]. A visual comparison between raw and CLAHE-enhanced images is shown in Fig. 3. Furthermore, geometric data augmentation is optionally enabled during training to assess its effect on model generalization. These preprocessing options are controlled via configuration and do not alter the underlying data splits or model architectures.



Fig. 3: Raw vs CLAHE contrast enhancement

Input Signals and Auxiliary Information. After preprocessing, each CNN model receives the full hand radiograph as its primary input signal. All architectures operate on the complete image and learn hierarchical feature representations directly from pixel intensities, capturing both global skeletal structure and localized morphological patterns relevant to skeletal maturation.

In addition to image data, patient gender is available as auxiliary metadata. This binary attribute is optionally incorporated into the learning framework via a learned embedding and concatenated with image derived features prior to regression. This design allows the influence of demographic information on predictive performance to be evaluated in a controlled manner, enabling direct comparison between models trained with and without gender input.

Dataset Distribution and Splits. The dataset is provided with predefined training, validation, and test partitions from the RSNA challenge:

- 12,611 images in the training set,
- 1,425 images in the validation set, and
- 200 images in the test set.

Bone age and gender distributions for each split are shown in Fig. 4. The bone age histograms exhibit a skew toward mid-adolescence, with the highest density occurring between approximately 11 and 14 years (130 – 170 months). Gender distributions are well balanced across all splits, with the training and validation sets containing approximately 54% male and 46% female subjects, and the test set evenly balanced.

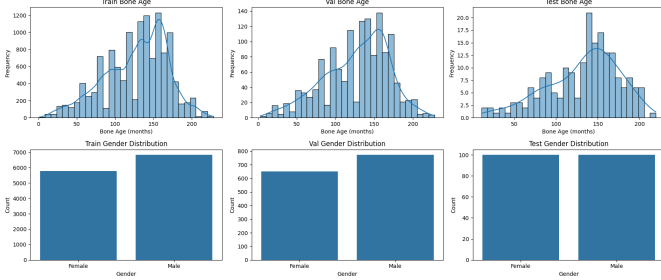


Fig. 4: Dataset distributions by split

The similarity of distributions across the training, validation, and test sets indicates that the predefined RSNA splits are statistically consistent. This reduces the risk of split-induced bias and supports fair comparison between different architectural configurations and input settings.

V. LEARNING FRAMEWORK

This section describes the convolutional neural network architectures, training strategy, and optimization procedures employed for bone age regression. All models are trained from scratch and use the same training protocol to allow fair comparison between architectures.

A. Baseline CNN

The Baseline CNN serves as a reference model and is designed to capture hierarchical skeletal features from the full hand radiograph. The architecture is loosely inspired by the VGG family of networks [15], adopting a sequential stack of convolutional layers with small receptive fields and progressive spatial downsampling.

Convolutional blocks. The network is composed of three convolutional blocks. Each block consists of two 3×3 convolutional layers followed by batch normalization and ReLU activation, and is concluded with a max-pooling operation. The number of filters increases across blocks ($32 \rightarrow 64 \rightarrow 128$), enabling the model to learn increasingly abstract feature representations as spatial resolution decreases. After the final block, a global average pooling layer aggregates the spatial feature maps into a 128-dimensional feature vector describing the overall skeletal morphology of the hand.

Regression head and gender integration. When gender information is enabled, the binary gender attribute (female = 0, male = 1) is processed through a learned embedding layer of dimension 8. The resulting embedding is flattened and

concatenated with the image-derived feature vector, forming a joint representation that is passed to the regression head. The resulting representation is passed through dropout regularization (rate 0.2) and a fully connected layer with 128 units and ReLU activation. A final linear output neuron produces the predicted bone age in months.

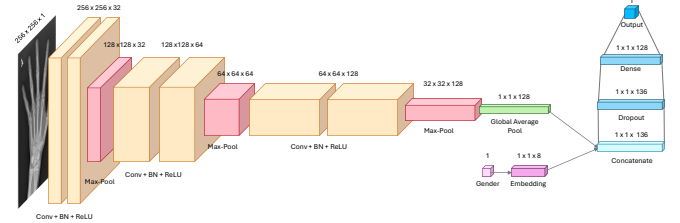


Fig. 5: Base CNN model architecture

This baseline architecture provides a compact and interpretable reference point for evaluating the benefits of deeper networks, residual connections, and multi-scale feature extraction. The model architecture is illustrated in Fig.5.

B. Skip Connection (SkipCon) CNN

The Skip Connection CNN extends the baseline architecture by incorporating residual learning mechanisms inspired by the ResNet [11] design paradigm. The model is designed to improve optimization stability and representational efficiency by enabling identity-based information flow across convolutional stages.

Stem. The network begins with a stem composed of a 7×7 convolution with stride 2, followed by batch normalization, ReLU activation, and a max-pooling layer with stride 2. This stem performs initial feature extraction while reducing the spatial resolution of the input radiograph.

Residual blocks. Following the stem, the architecture is organized into three residual stages, each consisting of two residual blocks. The number of feature channels increases across stages according to ($32 \rightarrow 64 \rightarrow 128$). Within each stage, residual blocks preserve spatial resolution, while spatial downsampling is performed at the transition between stages. Specifically, the first block of each stage beyond the first applies a stride of 2 in its initial convolution, reducing spatial resolution while increasing channel depth.

Each residual block comprises two 3×3 convolutional layers with batch normalization and ReLU activation. When spatial resolution or channel dimensionality changes, the shortcut connection is adapted using a 1×1 convolution with matching stride, ensuring proper alignment between the residual and shortcut paths. The output of each block is obtained via element-wise addition followed by ReLU activation. After the final residual stage, a global average pooling layer aggregates spatial feature maps into a $(1 \times 1 \times 128)$ feature vector.

Regression head and gender integration. When gender information is enabled, the binary gender attribute is processed through a learned embedding layer of dimension 8. The flattened embedding is concatenated with the image-derived

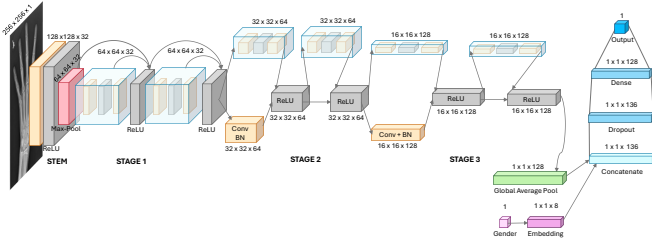


Fig. 6: Skip Connection (SkipCon) CNN model architecture

feature vector and forwarded to the regression head. The resulting representation is passed through dropout regularization (rate 0.2) and a fully connected layer with 128 units and ReLU activation. A final linear output neuron produces the predicted bone age in months. The complete architecture is illustrated in Fig. 6.

By introducing strided residual blocks for stage-wise downsampling, the SkipCon CNN combines hierarchical feature extraction with efficient gradient propagation. This design enables direct comparison with the baseline CNN to assess the impact of residual learning and deeper stage-wise representations on bone age regression performance.

C. Inception-based CNN (Inception CNN)

The Inception CNN is designed to exploit multi-scale feature extraction for bone age regression and is inspired by the Inception-V4 [12] architecture. Unlike purely sequential CNNs, Inception-style networks process feature maps through multiple parallel branches with different receptive fields, enabling the model to capture both fine ossification details and broader skeletal context within the same layer.

Stem. The network begins with a stem that progressively reduces spatial resolution while increasing channel depth. The input image is first processed by a stride-2 convolution followed by additional convolutional layers. Downsampling is then achieved through parallel branches consisting of max-pooling and stride-2 convolutions, whose outputs are concatenated. A second factorized downsampling step is applied in a similar manner. This stem design follows the Inception-V4 principle of performing early downsampling through parallel paths rather than relying on a single pooling operation.

Inception and reduction modules. After the stem, the architecture is organized into three groups of Inception modules operating at progressively lower spatial resolutions. In the configuration used in this work, the network includes: (i) two Inception-A blocks, (ii) three Inception-B blocks, and (iii) one Inception-C block. Between these groups, two dedicated reduction modules (Reduction-A and Reduction-B) perform spatial downsampling while increasing representational capacity.

Each Inception-A block consists of four parallel branches whose outputs are concatenated along the channel dimension: a 1×1 convolution branch, a $1 \times 1 \rightarrow 3 \times 3$ branch, a deeper $1 \times 1 \rightarrow 3 \times 3 \rightarrow 3 \times 3$ branch, and an average-pooling branch followed by a 1×1 projection. The subsequent Reduction-A

module performs spatial downsampling using a combination of stride-2 convolutional paths and a max-pooling path.

The Inception-B blocks operate on lower resolution feature maps and employ factorized convolutions with asymmetric kernels of 1×7 and 7×1 to increase receptive field size with reduced computational cost. Similar to Inception-A, each block includes multiple convolutional branches and an average-pooling projection branch, followed by concatenation. The subsequent Reduction-B module performs further downsampling using a combination of stride-2 convolutional paths (including factorized convolutions) and a max-pooling path.

Finally, the Inception-C blocks operate on the lowest-resolution grid and include split convolutions that branch into 1×3 and 3×1 paths to enrich multi-scale representation capacity.

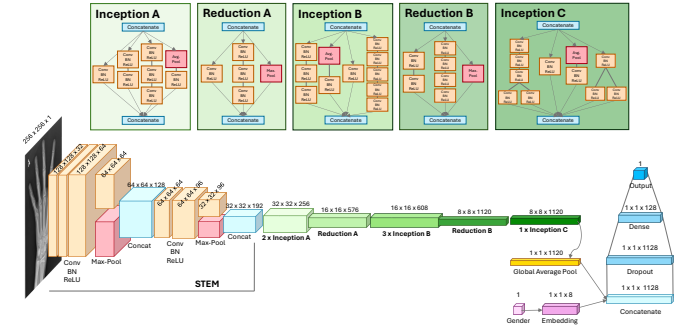


Fig. 7: Inception CNN model architecture

Regression head and gender integration. After the final Inception-C block, global average pooling aggregates the spatial feature maps into a fixed-length feature vector. When gender is enabled, the binary gender attribute is mapped through an embedding layer of dimension 8, flattened, and concatenated with the image-derived feature vector. The resulting representation is passed through dropout regularization (rate 0.2) and a fully connected layer with 128 units and ReLU activation. A final linear output neuron produces the predicted bone age in months. The overall structure is illustrated in Fig. 7.

This architecture enables evaluation of whether multi-branch, multi-scale feature extraction provides advantages over sequential and residual CNN designs for bone age regression.

D. Inception-Skip Connection (InSkipCon) CNN

The InSkipCon CNN combines multi-scale feature extraction with residual learning and is inspired by the Inception-ResNet-V2 [12] architecture. The model employs Inception-style parallel convolutional branches to capture patterns at different receptive fields, while using residual (skip) connections to improve optimization stability and enable feature refinement through residual updates.

Stem. The network begins with a stem that performs early feature extraction and spatial downsampling through a mixture of strided convolutions and pooling operations. Following

initial convolutions, downsampling is implemented via parallel branches (max-pooling and strided convolutions) whose outputs are concatenated, increasing channel capacity while reducing spatial resolution. This design follows the Inception family approach of achieving downsampling through parallel paths and concatenation rather than a single sequential pooling operation. The stem output forms the input to the subsequent Inception-ResNet stages.

Inception-ResNet modules and reduction blocks. After the stem, the architecture is organized into three groups of hybrid Inception-ResNet modules operating at progressively lower spatial resolutions. In the configuration used in this work, the network includes: (i) two Inception-ResNet-A blocks, (ii) three Inception-ResNet-B blocks, and (iii) one Inception-ResNet-C block. Between these groups, two reduction modules (Reduction-A and Reduction-B) perform spatial downsampling and channel expansion. As illustrated in Fig. 8, the feature map resolution decreases from 32×32 after the stem, to 16×16 after Reduction-A, and to 8×8 after Reduction-B, while the channel depth increases accordingly.

Each Inception-ResNet block computes a residual branch using multiple parallel convolutional paths. The outputs of these paths are concatenated and projected back to the shortcut dimensionality using a 1×1 convolution. The projected residual is then scaled and added to the shortcut connection, followed by ReLU activation. In this implementation, residual scaling factors are used to control the magnitude of the residual update ($\alpha_A = 0.17$, $\alpha_B = 0.10$, $\alpha_C = 0.20$), improving training stability in deeper multi-branch architectures. The reduction blocks follow the standard Inception reduction principle: multiple strided convolution paths and a pooling path are combined via concatenation to perform downsampling while preserving representational richness.

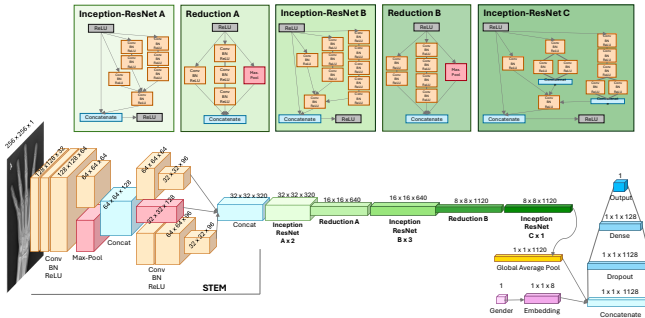


Fig. 8: Inception-Skip Connection (InSkipCon) CNN model

Regression head and gender integration. After the final Inception-ResNet-C block, global average pooling aggregates the spatial feature maps into a fixed-length feature vector. When gender information is enabled, the binary gender attribute is mapped through a learned embedding layer of dimension 8, flattened, and concatenated with the image-derived feature vector. The resulting representation is passed through dropout regularization (rate 0.2) and a fully connected layer with 128 units and ReLU activation. A final linear

neuron outputs the predicted bone age in months. The model structure is illustrated in Fig. 8.

Overall, the InSkipCon CNN represents the most expressive architecture considered in this study, integrating multi-scale processing and residual learning within a unified regression framework. This design allows assessment of whether combining Inception-style branching with residual connections provides additional benefits over the standalone Inception CNN and the ResNet-inspired SkipCon CNN under the same training and evaluation protocol.

E. Loss Function

Bone age estimation is formulated as a regression problem. All models are trained using the Huber loss [19] (also known as the smooth L1 loss), with a threshold parameter of $\delta = 10$. The Huber loss combines the desirable properties of mean squared error (MSE) and mean absolute error (MAE) by behaving quadratically for small errors and linearly for larger errors. For a prediction $\hat{y}$ and ground-truth bone age y , the Huber loss is defined as:

$$\mathcal{L}_\delta(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2, & \text{if } |y - \hat{y}| \leq \delta, \\ \delta |y - \hat{y}| - \frac{1}{2}\delta^2, & \text{otherwise.} \end{cases}$$

Model performance is evaluated using mean absolute error (MAE) and root mean squared error (RMSE), which are computed as:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|,$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2},$$

where N denotes the number of samples, y_i the ground-truth bone age, and $\hat{y}_i$ the predicted value for the i -th sample. MAE directly reflects the average prediction error in months, while RMSE penalizes larger errors more strongly, providing a complementary measure of model stability.

The Huber loss is selected as the training objective because it offers a more stable optimization landscape than MAE alone. While MAE is robust to outliers, its constant gradient magnitude and non-differentiability at zero can slow convergence. In contrast, the Huber loss provides smooth gradients for small errors, facilitating efficient learning during early and intermediate training phases, while retaining robustness to outliers through its linear behavior for larger errors.

F. Training Setup

All models are implemented in Python using the TensorFlow and Keras libraries. Training is performed using the Adam optimizer with an initial learning rate of 10^{-3} . Each model is trained for a maximum of 70 epochs, with early stopping employed to reduce overfitting. Early stopping monitors the validation MAE and terminates training if no improvement is observed for 10 consecutive epochs, restoring the model weights corresponding to the best validation performance.

To further improve optimization stability, the Keras ReduceLROnPlateau learning-rate scheduler is applied. When the validation MAE does not improve for 3 consecutive epochs, the learning rate is reduced by a factor of 0.2. This adaptive strategy allows larger updates during early training while enabling finer adjustments as the models approach convergence.

All experiments are conducted on a Windows system using the Windows Subsystem for Linux (WSL), with training accelerated by an NVIDIA GeForce RTX 3050 GPU.

Due to the substantially higher memory consumption of the InSkipCon CNN, a smaller batch size of 8 is used for this model. All other architectures are trained with a batch size of 16. This adjustment is made solely to accommodate GPU memory constraints and does not otherwise affect the training configuration.

VI. RESULTS

All models are evaluated using mean absolute error (MAE) and root mean squared error (RMSE), reported in months. Experiments are conducted using the predefined training, validation, and test splits of the RSNAPediatric Bone Age dataset. Model selection is performed based on validation MAE, and the corresponding model checkpoint is evaluated on the test set. Lower values of MAE and RMSE indicate better predictive performance.

A. Architecture Comparison Under the Best Configuration

Table 1 compares the four architectures under the best overall configuration identified in this study, i.e. Augment=False, CLAHE=True, Gender=True. Under this setting, the InSkipCon CNN achieves the lowest validation and test errors, with a test MAE of 6.74 months. The Inception CNN attains the second-best test MAE of 8.04 months, while the Base CNN and SkipCon CNN exhibit relatively higher errors.

TABLE 1: Model comparison under the best overall configuration (Augment=False, CLAHE=True, Gender=True).

	Base	SkipCon	Inception	InSkipCon
Val MAE	8.581	9.116	8.599	8.398
Val RMSE	11.762	12.169	11.537	11.173
Test MAE	8.870	8.935	8.035	6.741
Test RMSE	12.992	11.757	12.623	8.936
Params(M)	0.305	0.716	9.370	11.607
Train (h)	5.405	3.476	3.406	4.405

Overall, the results suggest that multi-scale feature extraction plays a key role in improving accuracy for bone age regression, and the best performance is obtained when Inception-style multi-branch processing is combined with residual learning. The relatively weak performance of the SkipCon CNN further indicates that adding residual connections alone is not sufficient to guarantee improved accuracy in this setting.

In addition to predictive performance, Table 1 highlights notable differences in computational behavior. Although the Base CNN has the smallest parameter count ($\approx 0.3M$), it exhibits the longest total training time of 5.4 hours under this configuration. Conversely, the Inception-based models complete training in less time despite having substantially more parameters (9–11M). This suggests that training duration is influenced not only by model size but also by convergence dynamics: architectures with stronger gradient flow and richer feature representations can reach their best validation performance in fewer effective training iterations, while simpler sequential models may require longer training to achieve comparable error levels.

It is important to note that this runtime pattern is specific to the configuration in Table 1. When considering all experimental runs, different trends emerge: the SkipCon CNN often achieves the shortest training times overall, whereas the InSkipCon CNN tends to be the most time-consuming due to its depth and higher memory footprint. The Base CNN and Inception CNN typically fall between these extremes. These observations reinforce that training time depends on multiple factors, including architecture, optimization behavior, and early stopping.

B. Results Across All Configurations

While Table 1 focuses on the best-performing configuration, the complete set of experiments includes an ablation study over three binary factors: data augmentation, CLAHE contrast enhancement, and gender as an auxiliary input. Table 2 summarizes the results for all combinations across the four architectures, reporting training/validation/test performance together with model size and training time. This view enables analysis of how architectural choices and preprocessing decisions interact and how they affect both predictive performance and computational cost.

Effect of Gender Input A consistent and substantial improvement is observed when gender is included as an auxiliary input. Across all four architectures and most preprocessing settings, enabling gender reduces both validation and test MAE. This trend is most visible for the Base CNN and

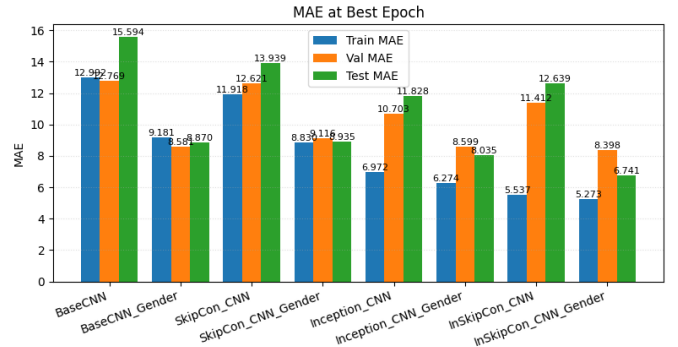


Fig. 9: MAE comparison across train/validation/test splits for each architecture trained with and without gender under Augment=False, CLAHE=True.

SkipCon CNN models, where the performance gap between gender-enabled and gender-disabled training is large. A plausible explanation is that male and female skeletal maturation trajectories differ systematically, providing gender helps the regression head learn a better mapping, reducing ambiguity in cases where visually similar growth patterns correspond to different chronological ages.

To illustrate this effect more clearly, Fig. 9 compares the MAE across training, validation, and test splits for each architecture trained with and without gender under the best preprocessing setup (Augment=False, CLAHE=True). The figure shows that gender consistently lowers validation and test MAE for all models. The improvement is particularly large for the lower-capacity Base CNN and SkipCon CNN, indicating that auxiliary metadata can compensate for limited representational power. In contrast, the Inception and InSkipCon models exhibit smaller but still consistent gains, suggesting that these architectures can extract stronger maturity related visual cues directly from the radiographs while still benefiting from demographic info.

TABLE 2: Complete results for all models and training configurations. A/C/G indicate Augmentation, CLAHE, and Gender usage (T=True, F=False). MAE is reported in months.

A	C	G	Model	Params(M)	Time(h)	Tr MAE	Val MAE	Test MAE
F	F	F	Base	0.30	2.79	12.62	12.07	13.37
F	F	F	SkipCon	0.71	0.74	13.02	12.55	14.71
F	F	F	Inception	9.37	3.00	4.53	11.64	12.50
F	F	F	InSkipCon	11.61	5.37	4.39	12.28	13.62
T	F	F	Base	0.30	1.98	12.88	13.51	19.05
T	F	F	SkipCon	0.71	1.49	12.57	14.20	19.02
T	F	F	Inception	9.37	3.21	8.14	12.23	14.53
T	F	F	InSkipCon	11.61	7.21	4.32	13.30	16.54
F	T	F	Base	0.30	5.42	12.99	12.77	15.59
F	T	F	SkipCon	0.71	4.84	11.92	12.62	13.94
F	T	F	Inception	9.37	5.50	6.97	10.70	11.83
F	T	F	InSkipCon	11.61	5.45	5.54	11.41	12.64
T	T	F	Base	0.30	6.36	12.86	12.54	14.59
T	T	F	SkipCon	0.71	2.72	13.41	13.15	15.57
T	T	F	Inception	9.37	6.40	6.17	12.22	15.42
T	T	F	InSkipCon	11.61	7.20	3.42	11.86	12.94
F	F	T	Base	0.31	2.74	10.22	9.68	11.31
F	F	T	SkipCon	0.72	0.65	10.01	9.81	9.54
F	F	T	Inception	9.37	2.29	6.01	8.73	8.30
F	F	T	InSkipCon	11.61	4.08	5.57	8.91	8.08
T	F	T	Base	0.31	2.01	9.03	8.93	8.52
T	F	T	SkipCon	0.72	0.82	10.29	10.32	10.98
T	F	T	Inception	9.37	1.69	7.70	9.11	8.92
T	F	T	InSkipCon	11.61	3.04	6.94	9.10	8.77
F	T	T	Base	0.31	5.40	9.18	8.58	8.87
F	T	T	SkipCon	0.72	3.48	8.83	9.12	8.94
F	T	T	Inception	9.37	3.41	6.27	8.60	8.03
F	T	T	InSkipCon	11.61	4.40	5.27	8.40	6.74
T	T	T	Base	0.31	4.74	10.76	10.30	10.87
T	T	T	SkipCon	0.72	3.47	9.39	9.94	9.72
T	T	T	Inception	9.37	4.02	6.14	8.85	8.04
T	T	T	InSkipCon	11.61	4.65	6.01	8.88	7.56

The magnitude of the improvement is smaller but still meaningful for the Inception CNN and InSkipCon CNN architectures. These models appear less dependent on demographic cues, likely because their higher representational capacity enables them to extract stronger maturity-related features directly from radiographs. Nevertheless, even for

these architectures, gender remains beneficial in the best-performing setting, and the overall best results of this study is obtained with gender enabled.

Effect of CLAHE Contrast Enhancement Table 2 shows that enabling CLAHE tends to improve performance, particularly when gender is also used. The benefit of CLAHE is most visible for the simpler models, where local contrast enhancement likely improves the visibility of bone boundaries and growth plate regions, making learning easier for limited capacity backbones. For the Inception-based architectures, the gains from CLAHE are often smaller in terms of MAE, which suggests that multi-scale feature extraction can partially compensate for contrast variability by learning robust representations. However, CLAHE provides an additional benefit for these higher-capacity models: it tends to improve training stability and reduce overfitting. In runs without CLAHE, the Inception-style models achieve very low training MAE while validation/test MAE remain noticeably higher, indicating a larger generalization gap. Enabling CLAHE narrows this gap by standardizing local contrast, effectively acting as a form of input-level regularization that improves generalization even when the absolute MAE improvement is modest.

Overall, these results indicate that CLAHE is a useful preprocessing option for this task, especially when computational constraints motivate smaller architectures. However, its impact is not uniform across all settings, which justifies treating it as a tunable preprocessing choice rather than a mandatory step.

Effect of Data Augmentation The effect of geometric data augmentation is mixed and, in several configurations, detrimental. In particular, under gender-enabled settings, augmentation often increases test MAE for the SkipCon and InSkipCon models. One possible explanation is that the RSNA dataset is relatively large and already contains natural variability across patients and acquisition conditions, which may reduce the need for strong augmentation. Taken together, the results suggest that mild augmentation may be acceptable, but aggressive augmentation should be treated cautiously in bone age regression, as it can degrade the fidelity of anatomical cues.

Training Efficiency and Early Stopping Behavior In addition to predictive accuracy, Table 2 highlights notable differences in training efficiency. Training time is not determined solely by the number of parameters but it also depends on convergence speed, early stopping, and the computational cost of the operations used in each architecture. In several runs, the SkipCon CNN reaches early stopping quickly and exhibits the shortest training times overall, whereas the InSkipCon CNN tends to require longer training and consistently has the highest computational cost due to its deeper multi-branch structure and greater memory demands.

Figure 10 visualizes the validation MAE trajectories during training and highlights clear differences in convergence behavior across architectures. Some models reach their best validation error in fewer epochs and stabilize earlier, while others improve more gradually. This helps explain why total training time is not determined by parameter count alone, models can

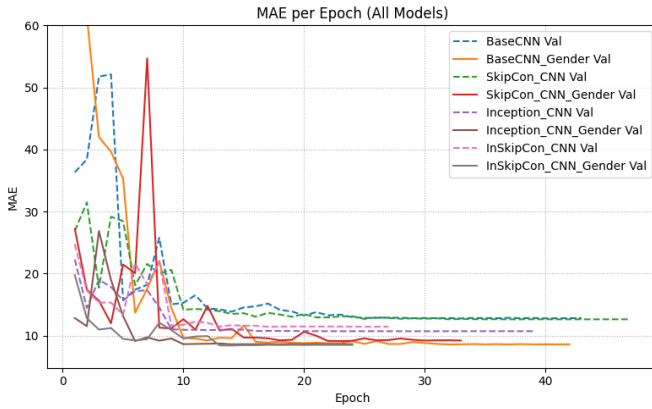


Fig. 10: Validation MAE as a function of epoch for all models trained with and without gender under Augment=False, CLAHE=True.

converge sooner even if they have more parameters.

The figure also shows that gender-enabled variants generally achieve a lower validation plateau than their gender-free counterparts, consistent with the error reductions observed in Fig. 9. In several cases, adding gender also leads to smoother validation curves, suggesting that the gender information reduces ambiguity in the regression mapping and can make optimization more stable.

C. Key Takeaways

Across all configurations, the InSkipCon CNN achieves the best overall test performance, while gender input consistently improves accuracy across architectures. CLAHE is generally beneficial, whereas augmentation does not provide consistent gains and can degrade performance depending on the model and setting. Training time varies with convergence and early stopping behavior, and is not determined by parameter count alone.

VII. CONCLUDING REMARKS

This project investigated convolutional neural network (CNN) approaches for pediatric bone age estimation from hand radiographs using the RSNA Pediatric Bone Age dataset. A complete, lightweight and reproducible pipeline was implemented in Python with TensorFlow and Keras libraries, including configurable preprocessing (CLAHE and data augmentation), optional gender integration, and a unified training setup. Four architectures were evaluated: a compact VGG-inspired Base CNN, a ResNet-inspired SkipCon CNN, an Inception-V4-inspired Inception CNN, and an Inception-ResNet-V2-inspired InSkipCon CNN. Across the explored settings, the best performance was obtained with Augment=False, CLAHE=True, and Gender=True, where the InSkipCon CNN achieved the lowest test error of 6.74 months.

The results highlight a practical accuracy-efficiency trade-off. InSkipCon provided the best predictive accuracy, but at the cost of substantially higher model complexity, memory usage, and training time compared to lighter architectures. At the same time, the experiments showed that training time

is not determined by parameter count alone and is strongly affected by convergence behavior and early stopping. The consistent gains from using gender suggest that incorporating simple demographic metadata can be a low-cost and effective way to improve regression performance, especially for smaller models. These findings are relevant for real-world deployment scenarios where hardware constraints, runtime budget, and interpretability requirements may dictate different model choices.

Future work could include stronger regularization and calibration strategies, and exploring region-aware designs (e.g., segmentation or attention-based localization) to improve robustness and interpretability. Additionally, a more systematic exploration of augmentation policies tailored to radiographic realism may help determine when augmentation is beneficial rather than harmful.

From a learning perspective, this project provided hands on experience in building end-to-end medical imaging pipelines, designing and comparing CNN families, and performing structured ablation studies. The main practical difficulty was balancing model complexity with limited GPU memory, which required adjustments such as reduced batch size for the InSkipCon CNN. Managing training stability, debugging pre-processing choices, and interpreting performance differences across configurations were also key challenges, but they helped reinforce the importance of careful experimental design and systematic evaluation.

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